

Managing Databases



Overview

- Database properties.
- Database expansion.
- Free space.
- Integrity checks.



Database Operations

- System Operation > Databases.

Databases

[Integrity Check](#)[Integrity Log](#)

Last update: 2020-04-01 15:39:27.702

To view details and to manage a local database, click the database name link from the table below:

Filter:

Page size:

Max rows:

Results: 10

Page: [<](#) [<<](#) **1** [>>](#) [>](#) of 1

☒ General view ☐ Free space view

	Name	Directory	Max Size (GB)	Size (MB)	Status	Encrypted	Journal
	IRISSYS	c:\intersystems\training\mgr\	Unlimited	90	Mounted/RW	No	Yes
	IRISLIB	c:\intersystems\training\mgr\irislib\	Unlimited	384	Mounted/R	No	No
	IRISTEMP	c:\intersystems\training\mgr\iristemp\	Unlimited	11	Mounted/RW	No	No
	IRISLOCALDATA	c:\intersystems\training\mgr\irislocaldata\	Unlimited	1	Mounted/RW	No	No
	IRISAUDIT	c:\intersystems\training\mgr\irisaudit\	Unlimited	1	Mounted/RW	No	Yes
	COMPANY	c:\intersystems\databases\company\	Unlimited	11	Mounted/RW	No	Yes
	CUSTOMER	c:\intersystems\databases\customer\	Unlimited	1	Mounted/RW	No	Yes
	ENSLIB	c:\intersystems\training\mgr\enslib\	Unlimited	163	Mounted/RW	No	No
	PERSONAL	c:\intersystems\databases\personal\	Unlimited	1	Mounted/RW	No	Yes
	USER	c:\intersystems\training\mgr\user\	Unlimited	1	Mounted/RW	No	Yes



Database Properties

- System Administration > Configuration > System Configuration > Local Databases.

Local Databases

Create New Database

Last update: 2020-04-01 15:40:58.524

The following is a list of the local databases:

Filter: Page size: Max rows: Results: 10 Page: of 1

Name	Directory	Size (MB)	Status	Resource	Encrypted	Journal		
IRISSYS	c:\intersystems\training\mgr\	90	Mounted/RW	%DB_IRISSYS	No	Yes	-	Globals
IRISLIB	c:\intersystems\training\mgr\irislib\	384	Mounted/R	%DB_IRISLIB	No	No	-	Globals
IRISTEMP	c:\intersystems\training\mgr\iristemp\	11	Mounted/RW	%DB_IRISTEMP	No	No	-	Globals
IRISLOCALDATA	c:\intersystems\training\mgr\irislocaldata\	1	Mounted/RW	%DB_IRISLOCALDATA	No	No	-	Globals
IRISAUDIT	c:\intersystems\training\mgr\irisaudit\	1	Mounted/RW	%DB_IRISAUDIT	No	Yes	-	Globals
COMPANY	c:\intersystems\databases\company\	11	Mounted/RW	%DB_%DEFAULT	No	Yes	Delete	Globals
CUSTOMER	c:\intersystems\databases\customer\	1	Mounted/RW	%DB_%DEFAULT	No	Yes	Delete	Globals
ENSLIB	c:\intersystems\training\mgr\enslib\	163	Mounted/RW	%DB_ENSLIB	No	No	Delete	Globals
PERSONAL	c:\intersystems\databases\personal\	1	Mounted/RW	%DB_%DEFAULT	No	Yes	Delete	Globals
USER	c:\intersystems\training\mgr\user\	1	Mounted/RW	%DB_USER	No	Yes	Delete	Globals



^DATABASE

```
TRAINING InterSystems IRIS TRM:1632
File Edit Help

%SYS>do ^DATABASE

1) Create a database
2) Edit a database
3) List databases
4) Delete a database
5) Mount a database
6) Dismount a database
7) Compact globals in a database
8) Show free space for a database
9) Show details for a database
10) Recreate a database
11) Manage database encryption
12) Return unused space for a database
13) Compact free space in a database
14) Defragment a database
15) Show background database tasks
16) Manage multi-volume database properties

option? █
```



Mounting Databases

- Startup mounts all databases marked required.
 - If required databases cannot mount, startup fails.
- Other databases are shown in unmounted state.
 - Will mount on first use.
- Mount/dismount manually using System Operation > Databases.

System > Configuration > Local Databases

Database Properties

Name	BILLDATA Required.		
Directory	/Host/SYS/mgr/user/		
Encrypted	No		
Block Size	8192 Bytes		
Size (MB)	Current 11 >=11	Expansion 0 0 for Default	Mo
Resource Name	%DB_USER Create New		
New Global Collation	IRIS standard		
Global Journal State	☒		
Preserve Global Attributes on Delete	☐		
Mount Read-Only	☐		
Mount Required at Startup	☐		
Stream Location			
New Volume Threshold Size	0	☐ TB	☐ GB ☒ MB



Database Maximum Sizes

Block Size	Maximum Database Size
8 KB (default, recommended)	32 TB
16 KB	64 TB
32 KB	128 TB
64 KB	256 TB

- Different block sizes yield different database maximum sizes.



Extended Database Size (v2026.1+)

- Allows new maximum database sizes that are 256x larger than current maximums.
 - Disabled by default.
 - Before enabling, run Integrity Check on existing databases.
 - Refer to later slides for more information on running Integrity Check.
- Once enabled, when database is within 100 MB of current limit, instance updates database format.
 - No scheduled downtime required.
 - Database will not mount on prior versions.



Extended Database Size (v2026.1+)

(cont.)

Block Size	New Maximum Database Size
8 KB (default, recommended)	8 PB
16 KB	16 PB
32 KB	32 PB
64 KB	64 PB

- Enable using System Administration > Configuration > Additional Settings > Compatibility > ExtendedDatabaseSize.
- Combined with Multivolume Database feature, provides sufficient control over database growth.
 - Single database can grow **very** large while contained in **multiple** OS-level files.



Database Expansion

- **Current.**
 - Immediately expands when higher value entered.
- **Expansion.**
 - Default expansion size (0) indicates expansion by larger of 10MB or 12% of current size (up to 1GB).
- **Maximum.**
 - Default (0, recommended) indicates:
 - For 8 KB: 8 PB (v2026.1+) or 32 TB (pre-v2026.1).
 - To prevent database from exceeding disk size.
 - Reaching Maximum causes <FILEFULL> error.
 - Does not freeze system.

Size (MB)	Current	765	Expansion	0	Maximum	0
		>=1		0 for Default		0 for Unlimited



Database Expansion (cont.)

- Disk drive must have enough room for IRIS.DAT to grow.
 - Regularly check OS file system for enough space.
- Notifications added to Messages log:
 - If free disk space low after expansion (not enough for next expansion).
 - If database reaching maximum size.



How To: Check Database Free Space

- System Operation > Databases > Free space view.
- do ^%FREECNT.
 - Run from %SYS to see for all databases.
- %SYS>do ^DATABASE.

Page: |< << **1** >> >| of 1 ☐ General view ☒ Free space view

Max Size	Size	%Available in existing file.	Available	% Free	Disk Free	
Unlimited			0.40MB	40	56.71GB	Mounted/RW
Unlimited	11MB	System Default	5.9MB	53.63	56.71GB	Mounted/RW
Unlimited	1MB	System Default	0.46MB	46	56.71GB	Mounted/RW
Unlimited	1MB	System Default	0.46MB	46	56.71GB	Mounted/RW
Unlimited			22MB	24.44	56.71GB	Mounted/RW
Unlimited			12MB	7.36	56.71GB	Mounted/RW

Amount of free disk space.

Space available in existing file.



Database Operations

- Mount, dismount, truncate, compact, defragment.
 - System Operation > Databases > Choose Database.

Database Details

MountDismountTruncateCompactDefragment

GENERAL INFORMATION		DATABASE SIZE	
Directory:	c:\intersystems\training\mgr\user\	Block Size:	
Resource Name:	%DB_USER	Blocks:	
Mounted:	Yes	Max Size:	
Read Only:	No	Size:	



Returning Empty Database Space to File System

- Step 1: Compact database.
 - Moves free space to end of IRIS.DAT file.
- Step 2: Truncate database.
 - Truncates empty space at end of IRIS.DAT file.
 - Also truncates .VOL files. (v2025.2+)
 - Truncated space returned to OS file system.
- Run compact and truncate commands:
 - Databases > Freespace.
 - %SYS>do ^DATABASE.
- **Recommended** to backup database beforehand and check integrity afterwards.



Returning IRISTEMP Space – By restart

- System Administration > Configuration > Additional Settings > Startup.
 - MaxIRISTempSizeAtStart controls truncation of IRISTEMP at startup.
- Truncate to appropriate starting size (MB).
 - Reduces database size after unexpected expansion.
 - No truncation if current size less than max.
 - 0 prevents truncation. (Default)



Returning IRISTEMP Space – Manually

- Manually truncate the database from the Management Portal if IRISTEMP grows too large



How To: Check Database Integrity

- System Operation > Databases > Integrity Check.
- %SYS>do ^Integrity
 - More options available in v2025.2+.
- Runs in background.
- Generates report on each database and global checked, including any errors.

This Integrity Check will be run in the background and the result will be saved in <install location>\mgr\integ.txt

☐ Stop after any error

Please check the databases that you want to perform the Integrity Check:

Name	Directory
☐ IRISYS	c:\intersystems\training\mgr\
☐ IRISLIB	c:\intersystems\training\mgr\irislib\
☐ IRISTEMP	c:\intersystems\training\mgr\iristemp\
☐ IRISLOCALDATA	c:\intersystems\training\mgr\irislocaldata\
☐ IRISAUDIT	c:\intersystems\training\mgr\irisaudit\
☐ BILLDATA	c:\intersystems\databases\billdata\
☐ COMPANY	c:\intersystems\databases\company\
☐ CUSTOMER	c:\intersystems\databases\customer\
☐ ENSLIB	c:\intersystems\training\mgr\enslib\
☐ PERSONAL	c:\intersystems\databases\personal\
☐ USER	c:\intersystems\training\mgr\user\

Select All

Unselect All

Select Globals

You may select specific globals for one selected database.

Run

Cancel



Integrity Check Recommendations

- Run at least once per week.
- Run in live environment when practical.
 - If integrity check takes too many hours:
 - Run on restored backup.
- Check integrity of backup.
 - Run integrity check on restored backup.



Managing Data and Code

- Viewing globals:
 - System Explorer > Globals > Select View for global.
 - `do ^%G`
- Deleting routines:
 - System Explorer > Routines > Delete Routines.
 - `do ^%RDELETE`
 - Delete all corresponding extensions.
 - .MAC, .INT and .OBJ.



Suggested Readings

- *System Administration Guide.*
 - *Managing InterSystems IRIS > Maintaining Local Databases.*



Summary

- What are the key points for this module?

